CoreSim: Interactive Nuclear Reaction Simulation Tool

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Purpose: To create a visually interactive simulator of a simplified nuclear reactor core, allowing users to control parameters (coolant flow, control rods, fuel type, etc.) and observe real-time physical behavior (like neutron flux, temperature, and power output).

Goal: To educate, engage, and demystify the science of nuclear fission reactors, particularly for students and science facilitators.

Target Audience: The target audience is a combination of high school/early college students interested in physics/engineering (mainly nuclear), teachers seeking interactive nuclear demos, and the general public interested in nuclear energy and reactors.

Features:

Feature	Description
Parameter Controls	Sliders for control rod %, coolant rate, fuel enrichment
Real-Time Feedback	Plots showing neutron flux, temperature, radiation, power
Visual Reactor Schematic	Graphical core that changes based on safety or status
Educational Popups	Context info explaining each variable and system
Challenge Mode?	“Safely reach 1GW without triggering a SCRAM”

Tools:
Language - Python
User Interface - Streamlit
Graphing - Matplotlib
Backend logic - NumPy/SciPy

Timeline:

Week	Deliverable
1	Wireframe + Brief
2–4	Physics simulation prototype
5–6	Frontend integration
7	Education layer
8	Launch & outreach

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|  🔧 Controls (left pane)        | 📊 Outputs (right pane)     |
|                                 |                             |
|  - Reactor Type (dropdown)      |  🔴 Status Indicator        |
|  - Control Rod % (slider)       |     (e.g., SAFE/MELTDOWN)   |
|  - Coolant Flow Rate (slider)   |                             |
|  - Fuel Enrichment (slider)     |  📈 Live Plot (Temp vs Time) |
|  - Power Goal (optional input)  |  📈 Live Plot (Flux vs Time) |
|                                 |                             |
|  ℹ️ Pop-up info buttons         |  ⬜ Core Diagram (optional)  |
|                                 |                             |
|                                 |  📚 "Explain This" Sidebar  |
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Component	Type	Why It Matters
Fuel Type	Input	Sets energy per fission & neutron yield
Enrichment	Input	Controls how easily fission chain starts
Control Rod %	Input	Primary method to manage neutron flux
Coolant Flow	Input	Controls temperature via heat removal

Neutron Flux	Output	Represents chain reaction strength
Reactivity	Derived	Shows if system is sub/critical/supercritical
Temperature	Output	Determines safety, core state
Power Output	Output	Shows usable reactor output
Status	Derived	Flags safe/unstable/meltdown conditions

Quantity	Equation	Importance
k_{eff}	From enrichment & rod %	Determines reactivity
ρ	$\frac{k-1}{k}$	Controls chain reaction speed
ϕ	$\phi \cdot (1+\rho) \cdot \gamma$	Neutron population
Power	$\phi \cdot E_f \cdot F$	Reactor output
Temp	$\Delta T = \frac{P \cdot \Delta t - Q_{\text{cool}}}{C}$	Determines safety

The Business Side (Ongoing): This is where you connect it to Econ. Create a one-page business plan using the "Business Model Canvas" framework. Outline your value proposition, customer segments, and revenue model (even if it's free). Track your metrics (users, feedback, etc.).

The Business Model Canvas

Designed for:	Designed by:	Date:	Version:

Key Partners

Who are our Key Partners?
Who are our key suppliers?
Which Key Resources are we acquiring from partners?
Which Key Activities do partners perform?

MOTIVATIONS FOR PARTNERSHIPS
Optimization and economy
Reduction of risk and uncertainty
Acquisition of particular resources and activities

Key Activities

What Key Activities do our Value Propositions require?
Our Distribution Channels?
Customer Relationships?
Revenue streams?

CATEGORIES
Production
Problem Solving
Platform/Network

Key Resources

What Key Resources do our Value Propositions require?
Our Distribution Channels? Customer Relationships?
Revenue Streams?

TYPES OF RESOURCES
Physical
Intellectual (brand patents, copyrights, data)
Human
Financial

Value Propositions

What value do we deliver to the customer?
Which one of our customer's problems are we helping to solve?
What bundles of products and services are we offering to each Customer Segment?
Which customer needs are we satisfying?

CHARACTERISTICS
Newness
Performance
Customization
"Getting the Job Done"
Design
Brand/Status
Price
Cost Reduction
Risk Reduction
Accessibility
Convenience/Usability

Customer Relationships

What type of relationship does each of our Customer Segments expect us to establish and maintain with them?
Which ones have we established?
How are they integrated with the rest of our business model?
How costly are they?

EXAMPLES
Personal assistance
Dedicated Personal Assistance
Self Service
Automated Services
Communities
Co-creation

Customer Segments

For whom are we creating value?
Who are our most important customers?

Mass Market
Niche Market
Segmented
Diversified
Multi-sided Platform

Channels

Through which Channels do our Customer Segments want to be reached?
How are we reaching them now?
How are our Channels integrated?
Which ones work best?
Which ones are most cost-efficient?
How are we integrating them with customer routines?

CHANNEL PHASES
1. *Awareness*
   How do we raise awareness about our company's products and services?
2. *Evaluation*
   How do we help customers evaluate our organization's Value Proposition?
3. *Purchase*
   How do we allow customers to purchase specific products and services?
4. *Delivery*
   How do we deliver a Value Proposition to customers?
5. *After sales*
   How do we provide post-purchase customer support?

Cost Structure

What are the most important costs inherent in our business model?
Which Key Resources are most expensive?
Which Key Activities are most expensive?

IS YOUR BUSINESS MORE
Cost Driven (leanest cost structure, low price value proposition, maximum automation, extensive outsourcing)
Value Driven (focused on value creation, premium value proposition)

SAMPLE CHARACTERISTICS
Fixed Costs (salaries, rents, utilities)
Variable costs
Economies of scale
Economies of scope

Revenue Streams

For what value are our customers really willing to pay?
For what do they currently pay?
How are they currently paying?
How would they prefer to pay?
How much does each Revenue Stream contribute to overall revenues?

TYPES
Asset sale
Usage fee
Subscription Fees
Lending/Renting/Leasing
Licensing
Brokerage fees
Advertising

FIXED PRICING
List Price
Product feature dependent
Customer segment dependent
Volume dependent

DYNAMIC PRICING
Negotiation (bargaining)
Yield Management
Real time Market

DESIGNED BY: Business Model Foundry AG
The makers of Business Model Generation and Strategyzer

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